Things to change:

Data and preprocessing. … For Q3, the same cleaned MCS panel was combined with monthly average EUR/CHF rates requested from the SNB data portal through its REST API with requests, and with the Grossregion holiday share built for Q2. …
Analytical approach. … Q3 estimates how strongly spending reacts to the EUR/CHF rate in border compared with non-border regions and then compares the border regions by neighbouring country (Section 4). …

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4 Q3: EUR/CHF exchange rate and border regions

This part of the project focused on testing a specific question: is spending in border regions more strongly affected by a decrease in EUR/CHF exchange rate than spending in non-border regions, and does this effect differ between border regions?

4.1 Method

Data acquisition.

During this phase monthly average EUR/CHF rates for 2019-2025 were requested from the SNB data portal via REST API (monthly average of 1 EUR in CHF) using requests; the received nested JSON answer was directly parsed into a pandas DataFrame and stored untouched as Parquet. A first request initially used yearly data returned only seven rows, too few to compute a regression, so the monthly series (84 rows) was used as a replacement.

Preprocessing.

Like the data acquisition, for the preprocessing part a marimo notebook was used; all columns of the Monitoring Consumption Switzerland retrieved data were checked for 9999 codes, missing and empty values, negative or zero values and duplicates. The results showed that only region and category contained 9999 (16,769 rows), which were removed, additionally unused category levels were dropped.
As a second step was decided to filter the data to match the time window with the one in scope for the project (2019-2025), the overall panel covers 2,557 days x 7 regions x 7 categories (125,293 rows), so no day was identified as "missing".
Part of the preprocessing was dedicated to outlier detection: outliers were flagged per region and category with the IQR rule, but kept as in scope, most of them are real events such as Christmas or extreme events such as lockdowns. In a brief EDA distributions were visualised, spending is right-skewed, so its log was used.
A COVID flag (16.03.2020 - 31.12.2022) and a border indicator were added together with calendar variables: as spending is only available by Grossregion, was defined that a region counts as a border region if it contains a canton with a main cross-border shopping area, i.e. Lake Geneva (GE; France), Northwestern (BS, BL; Germany), Eastern (SH, TG; Germany) and Ticino (Italy).

Merging and validation.

In this phase after a brief data quality check on retrieved exchange rates and holidays files, the first were joined by year and month followed by the Q2 holiday share by date and region (left joins). The holiday file only lists holiday dates, so the 116,179 rows without a holiday were set to 0. The DataFrame was then saved as Parquet.

Model.

Using statsmodels, log spending was regressed (OLS, HC1 standard errors) on the centred rate (1 unit = 0.01 CHF per EUR) and the interaction with the border indicator. The model controls for the spending level of each region-category combination, the weekday, the month, a yearly time trend (meaning that the year was entered as a single number, capturing the growth of card spending without absorbing the monthly exchange-rate effect), the COVID flag and the holiday share.
A second model replaces the border indicator with the neighbouring country. Nested models, variance inflation factors (VIF) and residuals were used to check to ensure the result does not depend on the choice of predictors, which was then re-estimated without the COVID period as a robustness check. Effects are expressed as % change in spending for the change of the rate from 2019 to 2025.

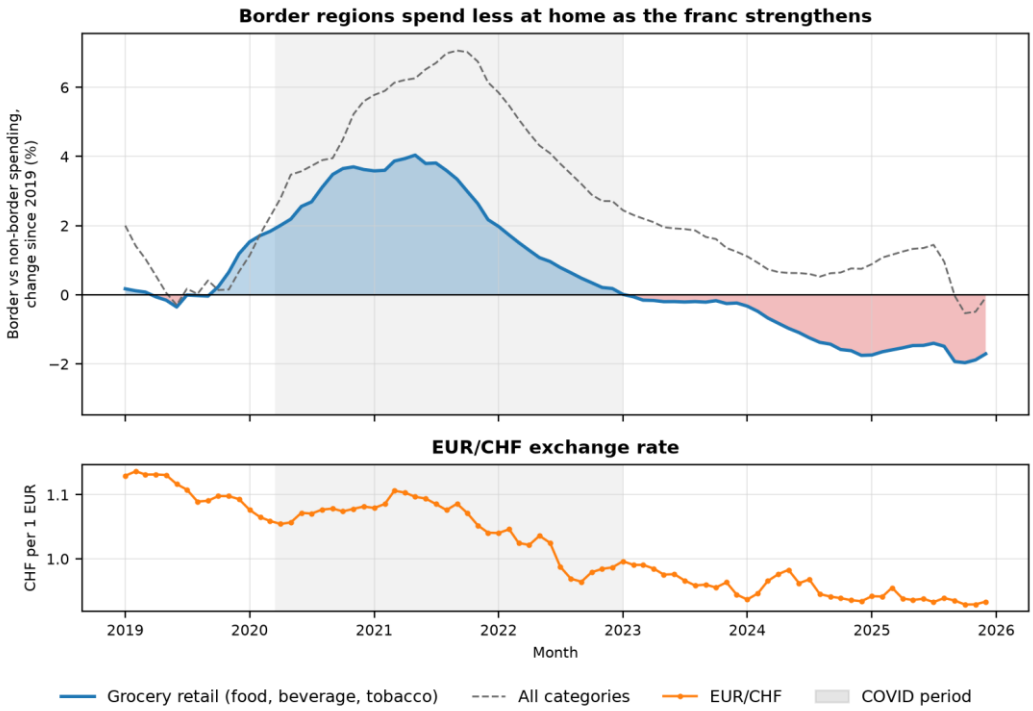


Figure 5. Spending in border relative to non-border regions in grocery retail (solid) and all categories (dashed), change since 2019 (12-month moving average, %), and monthly average EUR/CHF rate (SNB), 2019-2025. Grey: COVID period.

4.2 Results

- In grocery retail, spending in border regions moved with non-border regions in 2019, rose up to about 4% above them during the COVID period and fell below its 2019 relation from 2023, reaching around -2% in 2025, while EUR/CHF dropped below parity in 2022. All categories together show the same rise and decline but stay above zero until late 2025 (Figure 5).
- A stronger francs lowers spending more in border regions: the interaction is +0.34% per rappen, which corresponds to -5.8% for the 2019-2025 change. The coefficient is identical (0.0034) in all five nested specifications, while R^2 rises from 0.06 to 0.72 and the AIC falls from 403,964 to 252,919, so every group of controls improves the model without changing the result which validates the final model.
- The most impacted regions: the extra loss is largest around Lake Geneva (-13.7%), smaller in Northwestern and Eastern Switzerland (-3.5%) and not significant in Ticino (-2.2%, p=0.17). One important point: without the COVID period, the border effect shrinks to -1.6% and is no longer significant (p = 0.13).

Model	Observations	R^2	Border regions change [%]	p
Main model	125,293	0.72	-5.8% (-7.5 to -4.1)	<0.001
Without COVID period	75,264	0.78	-1.6% (-3.6 to +0.5)	0.13

Table 2. Extra change in spending in border vs non-border regions for the 2019-2025 change of EUR/CHF (-17.5 rappen)

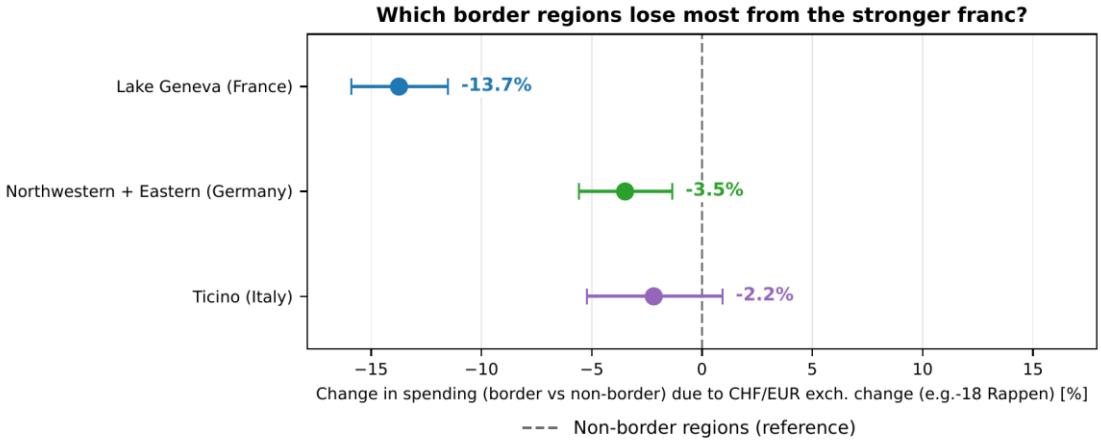


Figure 6. Extra change in daily scaled spending in each group of border regions compared with non-border regions (dashed line), for the change of EUR/CHF from 2019 to 2025 (-18 Rappen; % change, 95% confidence interval). OLS with region x category, weekday, month, year trend, COVID and holiday controls.

4.3 Discussion and limitations

- When the franc strengthens, spending falls more in border regions, consistent with residents shopping in the neighbouring EUR countries such as Germany, Italy or France. The effect is largest around Lake Geneva, which is almost surrounded by France; in Northwestern and Eastern Switzerland it is possibly diluted by areas far from the border (e.g. AG, GR, SG). In Ticino, a smaller price advantage or spending by cross-border commuters might offset it; the data unfortunately cannot test this.
- Much of the effect comes from the COVID period: closed borders kept shopping in Switzerland while the rate was still around 1.07-1.10 in 2020-2021, and the reopening coincided with the stronger franc with great synergy. If we exclude this period on its entirety the effect is not significant anymore, so it cannot be fully separated from the post-COVID normalisation.
- The border classification is at Grossregion level (Lake Geneva also includes VD and VS), with only four border and three non-border regions. As the rate falls almost steadily, its main effect is collinear with the year trend (VIF 12.5 and 10.5) and changes sign without the COVID period, so only differences between regions are interpreted. HC1 standard errors ignore correlation between days, so the intervals are probably too narrow. No causal effect can be assumed.